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Key Points:

- We use machine learning to quantify changes in Europe's recent heatwaves from incremental changes in cumulative anthropogenic emissions
- Despite decarbonization efforts, we find strong evidence that global emissions since 2015 have intensified Europe's recent heat extremes
- Regional-mean temperatures would be ~0.34°C cooler if 2025 heatwave weather conditions were to occur at 2015 levels of cumulative emissions

Supporting Information:

Supporting Information may be found in the online version of this article.

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Global Emissions Since the Paris Agreement Have Intensified Europe's Recent Heatwaves

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Abstract Understanding the consequences of continued greenhouse-gas emissions requires determining whether recent emissions can be robustly linked to changes in extreme weather events. However, current approaches have a limited ability to quantify changes in extreme weather caused by small subsets of anthropogenic emissions, particularly for individual events. To address these limitations, we train a generative machine-learning model to predict spatially-explicit changes in daily surface temperature at different levels of cumulative emissions, conditional on the event's observed meteorological conditions. Using this approach, we find strong evidence (>95% probability) that the combined effects of anthropogenic emissions released since the 2015 UN Paris Agreement have increased the intensity of Europe's summer temperature extremes since 2021. Additionally, our results suggest >99% probability that regional-mean temperature during Europe's high-impact June 2025 heatwave would have been lower (median estimate of 0.34°C) if the same large-scale meteorological conditions had occurred under 2015 levels of cumulative emissions.

Plain Language Summary It is well understood that, in aggregate, greenhouse gas emissions have increased the frequency and intensity of extreme weather events and their impacts. However, quantifying the effect of small increments of emissions on the intensity of specific weather events remains challenging. To overcome these challenges, we use machine learning techniques to predict changes in Europe's most intense heatwaves for subsets of cumulative anthropogenic emissions. We find strong evidence that relatively small fractions of global emissions, including those released since the 2015 UN Paris Agreement, have intensified recent heat events in Europe, including the June 2025 heatwave. This approach is an important step toward quantifying the contribution of small amounts of anthropogenic emissions to damages caused by individual extreme weather events, which is crucial for understanding the consequences of continued emissions during decarbonization.

1. Introduction

The 2015 UN Paris Agreement (UNFCCC, 2015) codified efforts to avoid the most damaging impacts of climate change, including those caused by extreme weather events (Seneviratne et al., 2021), through reductions in greenhouse gas emissions. Despite ongoing decarbonization efforts, global emissions continue to rise (Friedlingstein et al., 2025). Understanding the consequences for people and ecosystems requires robust quantification of how extreme weather events have responded to continued emissions in recent years.

Robust links have been found between historical global warming and changes in a wide variety of extreme weather events (Diffenbaugh et al., 2017; Kimutai et al., 2025; Peterson et al., 2012; Stott et al., 2004; Swain et al., 2020) and their impacts (Callahan et al., 2025a; Davenport et al., 2021; National Research Council, 2026; Perkins-Kirkpatrick et al., 2022; Strauss et al., 2021). These attribution studies are an important step toward assigning responsibility for climate damages (Noy et al., 2023; Wehner & Reed, 2022). However, questions remain about whether subsets of emissions (such as those released in recent years or by individual emitters) can be quantitatively linked to changes in extreme weather, particularly for individual extreme events (King et al., 2023).

A long-standing challenge facing extreme event attribution is that the signal of anthropogenic forcing can be small relative to the noise of natural climate variability (Trenberth et al., 2015). This challenge is exacerbated when attempting to robustly quantify the influence of a subset of historical emissions (for which the contribution to total

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climate forcing is particularly small) on individual extreme weather events (for which the role of natural variability is particularly pronounced). To mitigate this issue, recent studies have used probabilistic attribution techniques (Philip et al., 2020) applied to large climate model ensembles to quantify changes from specific amounts of cumulative carbon emissions (Otto et al., 2017; Pall et al., 2011), including emissions from the highest emitting entities (Callahan & Mankin, 2025; Nzotungicimpaye & Damon Matthews, 2024; Quilcaille et al., 2025). However, a key limitation of these probabilistic approaches is that, by analyzing changes within broad categories of events, they potentially ignore differences in the response to emissions between individual events within each category (Feser & Shepherd, 2025).

This sensitivity to specific spatiotemporal and atmospheric characteristics has motivated “event-specific” or “storyline” attribution approaches (Shepherd, 2016) that are grounded in the precise meteorological conditions of a given event (Bercos-Hickey et al., 2022; Patricola et al., 2022; Reed & Wehner, 2023). These storyline techniques excel in capturing the thermodynamic effect of global warming conditional on the meteorological conditions of the event (Feser & Shepherd, 2025), but they fail to account for dynamically-driven changes in event frequency (Faranda et al., 2023), and typically require large computational resources for thorough uncertainty quantification (Reed et al., 2020). Thus, despite recent advances, including those incorporating machine learning (Baño-Medina et al., 2025; Jiménez-Estevé et al., 2025; Trok et al., 2024), linking changes in individual extreme events to specific increments (or “shares”) of greenhouse gas emissions remains challenging.

We design a storyline approach which uses generative machine-learning to predict changes in daily surface temperature fields caused by the combined effects of anthropogenic forcing associated with incremental changes in cumulative global fossil fuel CO₂ emissions. After training on an ensemble of climate model simulations, we apply the learned relationships to quantify the influence of recent emissions on recent extreme events. Our framework uses diffusion models, which were initially developed for image generation (Ho et al., 2020) but have recently been applied to climate model downscaling and forecasting (Lopez-Gomez et al., 2025; Nai et al., 2024). Crucially, this approach allows us to efficiently generate multiple spatially-explicit predictions consistent with the specific meteorological conditions of each event. This is impractical with current attribution methods, but enables us to account for multiple uncertainties, including natural variability and differences between climate models.

To understand how recent extreme weather events have responded to continued greenhouse gas emissions, we apply this machine learning-based framework to quantify the influence of global anthropogenic emissions since the 2015 Paris Agreement on the frequency and intensity of Europe's recent summer heat extremes. This includes in-depth analysis of the influence of post-2015 emissions on Europe's early-summer 2025 heatwave (Guglielmi, 2025), which caused widespread damages and thousands of heat-related deaths—many of which have been attributed to global warming since the pre-industrial period (Barnes et al., 2025; Clarke et al., 2025).

2. Materials and Methods

Building on the framework of Trok et al. (2024), we design an event-specific approach to quantify changes in heatwave intensity caused by incremental changes in cumulative anthropogenic emissions, such as the total released since 2015. Using output from 10 global climate models (GCMs) (see Section 2.1), we train a diffusion model to predict daily surface temperature anomaly fields over Europe using several input variables, including daily meteorological fields and cumulative fossil CO₂ emissions (Figure S1 in Supporting Information S1). During training, we implement a denoising diffusion algorithm which allows us to generate conditional distributions of possible daily temperature fields which incorporate different sources of uncertainty (see Section 2.2). Using out-of-sample inputs from the ERA5 reanalysis, we then predict changes in the temperature of observed extreme events from post-2015 emissions (see Section 2.3).

2.1. Diffusion Model Input Data

The diffusion model uses the following inputs: 2-dimensional fields of daily sea-level pressure, top-layer soil moisture, and geopotential height from 700, 500, and 250 mb (which we refer to collectively as “daily meteorological fields”), as well as the cumulative fossil fuel CO₂ emissions since 1850, the calendar day, and a label indicating the GCM (Figure S1 in Supporting Information S1). Our regional domain extends across Europe and surrounding regions (−34° to 54°E and 21° to 75°N) at 2.81° horizontal resolution. We train three separate diffusion models using daily climate model output from 10 CMIP6 GCMs (see Table S1 in Supporting Information S1), which include natural and anthropogenic climate forcings from 1850 to 2014, but different shared

socioeconomic pathways (SSP2-4.5, SSP3-7.0, and SSP5-8.5) to project future anthropogenic emissions from 2015 to 2100. Although the CMIP6 high-forcing scenario SSP5-8.5 will be discontinued in future CMIP versions (Van Vuuren et al., 2026), these SSP5-8.5 simulations still provide a valuable source of training data, allowing us to compare results from diffusion models trained on a range of emissions. (See Text S1 in Supporting Information S1 for additional details.)

Aside from the direct effect of greenhouse gas and aerosol emissions on global radiative forcing, these emissions also cause long-term changes in—and feedbacks between—Earth System components that also influence regional temperature trends. To ensure that the diffusion model associates these long-term temperature trends with changes in the CO₂ emissions variable rather than trends in daily meteorological inputs (e.g., soil drying, thermal expansion of the geopotential height inputs), we remove the long-term calendar-day linear trends at each grid cell of the daily input fields calculated across all realizations with respect to the cumulative fossil fuel CO₂ emissions input variable (similar to Trok et al. (2024), which used global mean temperature as the explanatory variable). As a result of this detrending, the diffusion model allows us to quantify the combined effect of anthropogenic forcing on Europe's daily temperature (including secondary effects mediated through changes in the land and atmosphere) by making temperature predictions at different levels of cumulative fossil CO₂ inputs.

To construct the cumulative fossil fuel CO₂ emissions input, we use annual emissions data from the input4MIPs data set, which contains emission forcing files for the CMIP6 experiments. This data set includes fossil CO₂ emissions from all sources, but excludes land-use emissions and non-CO₂ forcing agents (which have limited availability for these GCM simulations). Consequently, for our main analysis, the relationship between temperature and cumulative fossil CO₂ emissions learned by the diffusion model reflects the combined effect of all forcing agents (including other greenhouse gas and aerosol emissions) that are correlated with a given share of cumulative fossil CO₂ emissions in the GCM simulations, though we also test the sensitivity of our results to non-CO₂ greenhouse gases and other non-greenhouse gas forcings. (See Text S1 in Supporting Information S1 for additional details).

2.2. Diffusion Model Design

Our diffusion model follows the common setup of a convolutional U-net neural network architecture, consisting of a descending or contracting path where dimensionality is reduced followed by an ascending or expanding path where the learned features are decoded into an image (see Figure S1 and Text S2 in Supporting Information S1 for details). We train a convolutional U-net neural network using a denoising diffusion algorithm (Ho et al., 2020). While there are many variations of diffusion algorithms, they all follow the same two-step framework: a simple forward process that adds noise to an image, and a reverse process that uses a trained neural network (i.e., “diffusion model”) to iteratively remove noise from that image over many steps. (See Text S3 in Supporting Information S1 for details).

We design our diffusion model to capture two key sources of uncertainty: (a) GCM differences in the global and regional climate response and (b) natural climate variability. For (a), we provide one-hot encoded GCM labels to each training sample (Callahan et al., 2025a, 2025b) which allow the diffusion model to optimize its predictions for each specific GCM (e.g., Figure S2a in Supporting Information S1). This design choice likely also reduces the sensitivity of the diffusion model behavior to the inclusion or exclusion of any one GCM, though a leave-one-out GCM test (e.g., Gordon & Diffenbaugh, 2025) would be needed to show this explicitly. For (b), during training the diffusion model learns to predict temperature distributions that are sufficiently wide to capture the range of possible surface temperatures that could occur for the same large-scale circulation conditions due to variations in fine-scale processes that influence temperature (land-atmosphere interactions, boundary-layer processes, etc.).

2.3. Estimating Changes in Heatwave Intensity Caused by Post-2015 Emissions

After training the diffusion model on GCM simulations, we predict changes in recent heatwaves due to emissions since 2015. We use the diffusion model to generate two distributions of possible heatwave temperatures: one for cumulative fossil CO₂ emissions at the time of the event, and one for cumulative emissions by 2015 (from Friedlingstein et al., 2025). We generate 40 counterfactual temperature predictions for each day at each 100 Gt increment of CO₂ emissions, using a different random seed for each prediction, and treating each daily timestep independently. Although temporal consistency is largely preserved through the daily meteorological input maps, we also test the sensitivity of our results to autocorrelated sampling (see Text S4 in Supporting Information S1).

Then, treating each GCM and SSP scenario separately, we calculate the difference between these two distributions by taking pairwise differences between temperature values from all possible combinations of non-matching random seeds (Figure S2; Text S4 in Supporting Information S1). We then combine these temperature differences from all GCM and SSP scenarios, yielding a single distribution of attributable temperature change for the event.

For the out-of-sample predictions using the specific meteorological conditions of each event, we obtain the necessary daily input data from version 5 of the ECMWF Reanalysis (ERA5) (Hersbach et al., 2023). To be consistent with the GCM training data, we detrend and standardize these ERA5 inputs at each grid cell to control for the mean response to climate forcing. (See Text S1 in Supporting Information S1 for details).

3. Results

3.1. Diffusion Model Evaluation

We first assess the model's ability to predict daily temperature for both the out-of-sample ERA5 data and unseen samples from the GCM training data set (Figure 1). Despite a cold bias at most individual grid cells (Figure 1a), our diffusion model predicts regional-mean ERA5 temperature with a mean absolute error (MAE) of 0.28°C, a bias of −0.08°C, and an average pattern correlation of $R = 0.89$ (daily values from $R = 0.67$ to $R = 0.96$) across all summer days from 1979 to 2025 (Figure 1b), with a larger cold bias but similar overall performance across the warmest 10% of summer days (MAE = 0.32°C, bias = −0.20°C, pattern correlation = 0.88). To assess the spread of the predicted regional temperature distribution, we compute the probability integral transform with 10 bins (Figure 1c) and find that the fraction of daily ERA5 values in each decile of the predicted distribution varies from 6% to 13%, with a larger fraction falling in the warm tail (consistent with a slight cold bias). If we remove the mean prediction bias (by adding +0.08°C to all predictions), we find that <10% of target values fall in the most extreme decile bins (Figure S3 in Supporting Information S1). This suggests that the diffusion model tends to overestimate daily temperature variability, meaning that our attribution uncertainty ranges are likely conservative (i.e., larger than they would be in a perfectly calibrated model). Compared to the ERA5 data, we find similar performance on the unseen GCM data set (MAE = 0.20°C, bias = −0.05°C; Figure 1b) and a similar spread across the predicted distribution for each individual GCM (Figure S4 in Supporting Information S1). Here, and throughout our analysis, mean predictions and uncertainty are based on 100 predictions generated for each day from 1979 to 2025, and 40 daily counterfactual predictions at each 100 Gt increment of fossil CO₂ emissions.

While it is challenging to verify attribution results for individual events (Diffenbaugh, 2020), evaluating the diffusion model's prediction of long-term temperature trends (Figures 1d–1i) provides a useful test of the model's ability to predict the response of daily temperature to changes in emissions. The model's predicted changes in annual maximum weekly temperature agree well with the magnitude and spatial pattern ($R = 0.99$) of GCM ensemble-mean trends (Figures 1d and 1e). Additionally, the GCM-specific regional trends in the annual hottest week are consistent with predicted trends corresponding to each GCM label input variable ($R = 0.98$) (Figure 1f), as are the summer-mean temperatures (Figure S5 in Supporting Information S1). We also find that predicted decadal trends (1979–2025) in the annual hottest week (Figure 1h) are generally consistent with ERA5 trends at each grid cell (Figure 1g), despite some inconsistencies over the Middle East and northern Africa. At the regional scale, the diffusion model's predictions capture the long-term trend (within 0.02°C per decade) and interannual variability in the ERA5 annual maximum weekly temperature time series (Figure 1i), which further builds confidence in predicting changes in individual extreme heatwaves at different levels of emissions.

3.2. Post-2015 Emissions Intensified Europe's Recent Summer Heat Extremes

After thoroughly evaluating the diffusion model's temperature predictions, we quantify the influence of post-2015 emissions on the intensity of the hottest week of the year from 2016 to 2025 (Figures 2a–2c). Our results suggest >95% probability that post-2015 emissions have increased regional-mean temperature during the hottest week for each individual year from 2021 to 2025 (Figures 2b and 2c). During 2021–2025, the predicted contribution of post-2015 emissions to heatwave intensity is more than 0.35°C (median estimate) for some individual grid cells, particularly in southern and eastern Europe where the magnitude of attributable change is largest (Figure 2a).

We extend this analysis to extreme heat frequency by analyzing changes in the annual number of days above the 95th percentile summertime temperature threshold (T95; relative to the first 30 years of the satellite era, 1979–2008), conditional on the same sequence of historical weather patterns from 2016 to 2025 recurring without

Evaluating predictions on out-of-sample ERA5 data for JJA days in 1979-2025

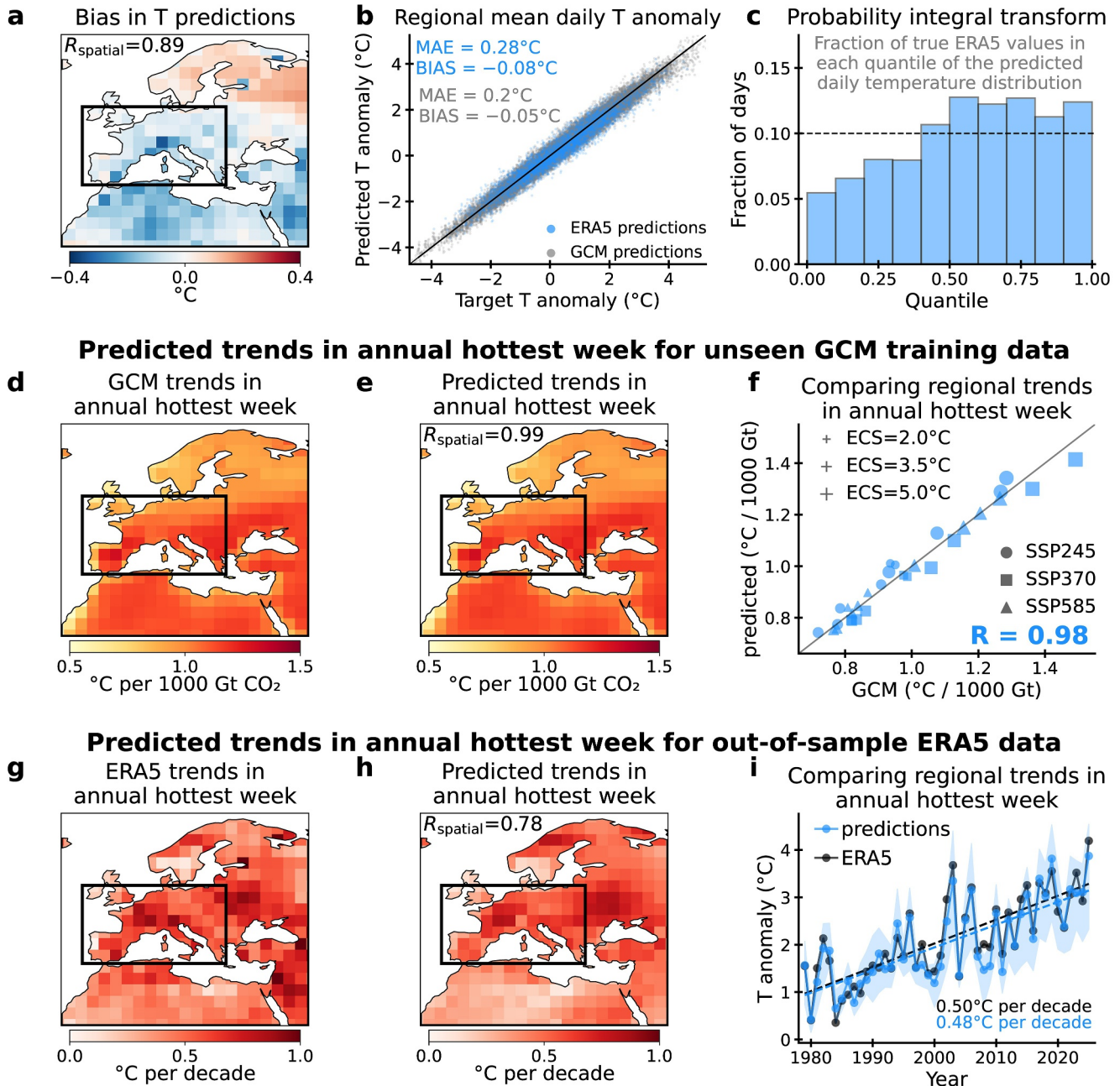


Figure 1. Evaluating diffusion model temperature (T) predictions. (a) Bias in T anomaly predictions for ERA5 data. (b) Predicted T and the target values for ERA5 (blue) and the unseen GCM data (gray) averaged across the black box in panel (a). (c) The fraction of daily T values that fall within each decile of the predicted daily T distribution. (d–f) Trends in the annual hottest week from (d) GCM ensemble-mean, (e) model predictions, (f) regional-mean GCM trends and predictions for each GCM input label and SSP scenario. Size in (f) indicates equilibrium climate sensitivity (ECS; Meehl et al., 2020) and shapes indicate SSP scenario. (g–i) Trends in annual maximum 7-day T for the 1979–2025 period from (g) ERA5, (h) model predictions, and (i) regional mean.

post-2015 emissions. We find >95% probability that post-2015 emissions have increased the annual number of days above that threshold for each individual year from 2022 to 2025 (Figures 2e and 2f). The increase exceeds 4 days per year for some individual grid cells, particularly near the Mediterranean where the magnitude of attributable change is largest (Figure 2d).

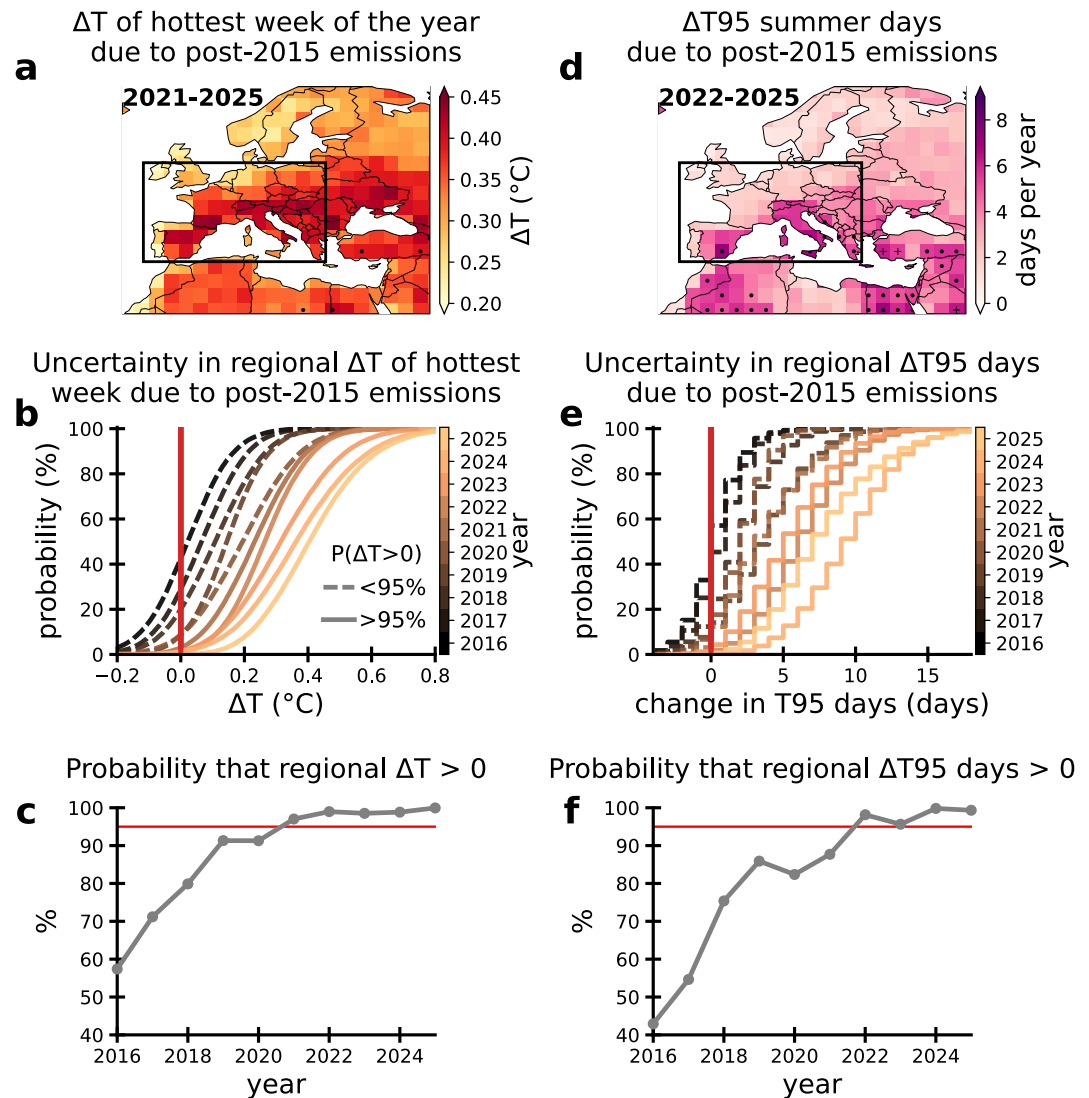


Figure 2. Post-2015 emissions intensified Europe's recent summer heat extremes. (a) Influence of post-2015 emissions on the hottest week of the year averaged across 2021–2025 (i.e., years with >95% probability that regional $\Delta T > 0$ from panel (c)). Dots (plus signs) indicate >95% probability that $\Delta T > 0$ for (more than) 1 year from 2021 to 2025. (b) Distribution of regional-mean changes in the hottest week of the year from 2016 to 2025. Solid lines indicate that the probability that $\Delta T > 0$ (shown in c) exceeds 95%. (d–f) Same as (a–c) but for the number of days above the JJA 95th-percentile threshold from 1979 to 2008.

3.3. Small Subsets of Emissions Intensified the 2025 European Heatwave

To understand the influence of post-2015 emissions on a single high-impact extreme event, we quantify changes in Europe's 2025 early-summer heatwave (June 24th to July 3rd) (Figure 3a). We find that the diffusion model predicts the observed spatial dipole pattern of temperature anomalies ($R = 0.90$), and the regionally-averaged median prediction (3.59°C) is within 10% of the ERA5 anomaly (3.98°C) (Figure 3b). In addition, the predicted regional-mean temperature distribution encompasses the ERA5 value for most individual days (Figure S6 in Supporting Information S1), and nearly all individual grid cells in the event region (Figure 3b, Figure S7 in Supporting Information S1). By taking the difference between predictions at 2025 (Figures 3b) and 2015 cumulative emissions (Figure 3c), we find that global emissions since 2015 intensified this event by $+0.25^{\circ}$ to $+0.50^{\circ}\text{C}$ depending on the grid cell, with the largest contributions in the Mediterranean region (Figure 3d). By comparing regional-mean temperature distributions at 2025 and 2015 levels of emissions (Figure 3e), we

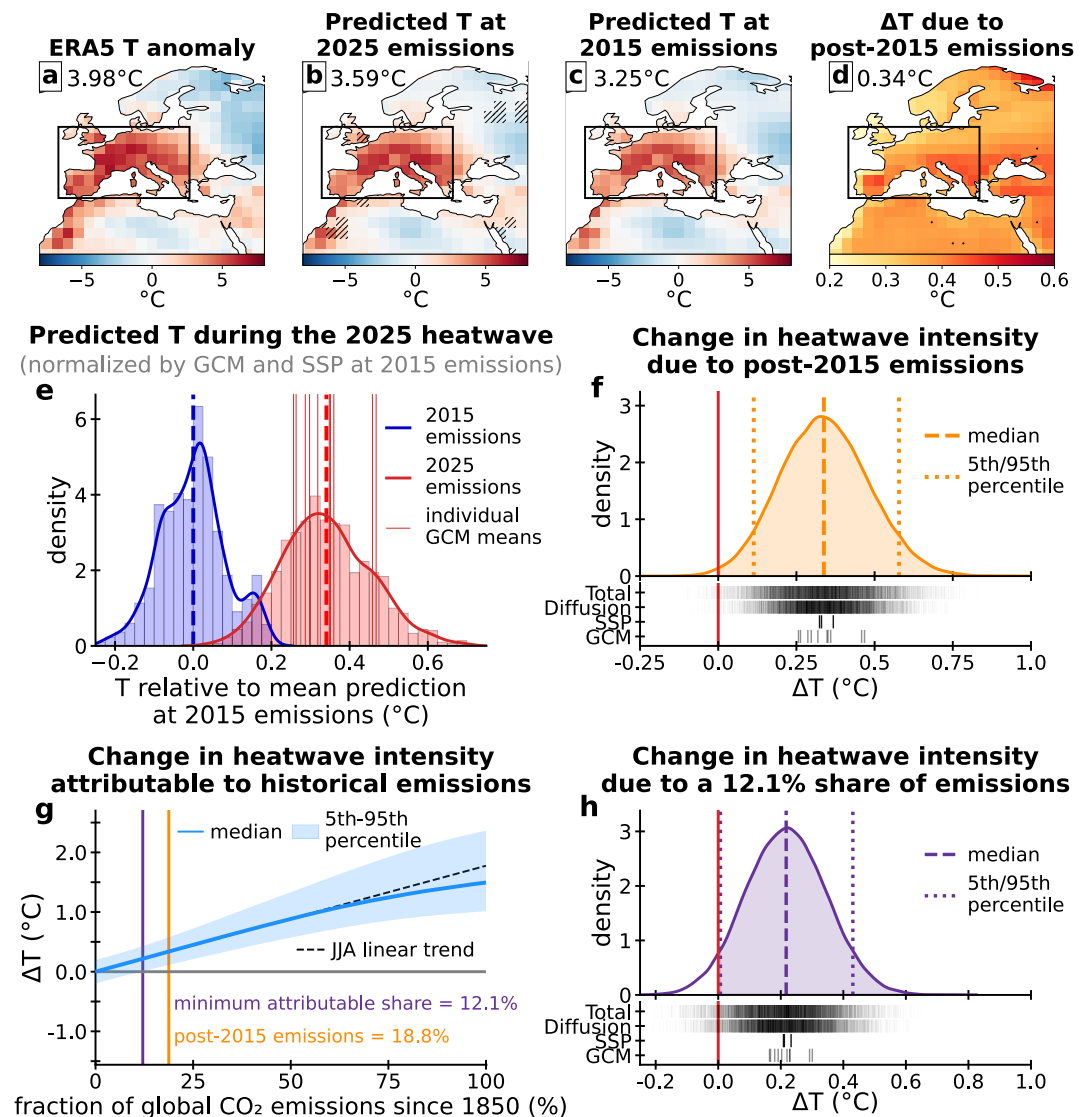


Figure 3. Post-2015 emissions intensified Europe's June 2025 heatwave. (a–d) Temperature (T) anomaly from June 24 to July 3rd for (a) ERA5, (b) prediction at 2025 emissions, (c) prediction at 2015 emissions, (d) difference between (b) and (c). Dots in (d) indicate >95% probability that $\Delta T > 0$. Text shows the regional mean (black box). Hatching in (b) indicates where the ERA5 value in (a) falls outside the predicted distribution. (e) Regional-mean T at 2015 (blue) and 2025 (red) levels of emissions (after removing the predicted mean for each GCM and SSP at 2015 emissions; see Figure S2 in Supporting Information S1). (f) Change in regional-mean T due to post-2015 emissions. (Bottom panel) Spread across all values while holding the other variables constant. (g) Predicted change in regional-mean temperature (ΔT) for a given percent change in total emissions since 1850 (relative to 2025 levels). Vertical lines indicate the fraction of total emissions released since 2015 (orange) and the minimum fraction for which the 5th percentile is $>0^\circ\text{C}$ (purple). Black line shows ensemble-mean GCM linear trend in June–August T . (h) Same as panel (f) but for a 12.1% share of global emissions since 1850.

calculate >99% probability that post-2015 emissions increased regional-mean temperature during the June 2025 heatwave, and 95% probability that the contribution of post-2015 emissions exceeded 0.11°C (Figure 3f).

We find that this estimated change in Europe's 2025 heatwave temperature (mean estimate of 0.34°C) due to 10 years of additional forcing is consistent with 10-year differences in annual heat events in the GCM simulations (mean of 0.42°C across 30 GCM simulations) (Figure S8 in Supporting Information S1). While the mean in these estimates is very similar, the uncertainty in the GCM analysis is an order of magnitude larger. This reduction in attribution uncertainty for quantifying the effect of relatively small increments of emissions on single events is a key advantage of the diffusion model approach. We also test the sensitivity to our treatment of additional

greenhouse gases and aerosols (Figure S9 in Supporting Information S1), and an alternative random sampling technique used to construct event time series with increased autocorrelation (Figure S10 in Supporting Information S1).

To generalize these results, we calculate changes in Europe's 2025 early-summer heatwave due to varying increments of cumulative historical emissions ranging from 0% to 100% of global emissions since 1850 (Figure 3g). For the framework presented here, a 12.1% share of global cumulative emissions since 1850 is the smallest increment for which we estimate >95% probability that emissions intensified the event (i.e., the attributable change in regional-mean temperature is >0°C) (Figure 3h). Although variability in the diffusion model predictions presents the largest source of uncertainty in these attribution estimates for small increments of emissions, we find that GCM differences become increasingly important for larger shares of cumulative global emissions, especially those exceeding 50% of the historical total (Figures S11c, S11f, and S11i in Supporting Information S1). We also find that the median response of event intensity to cumulative emissions can be slightly steeper at lower fractions of cumulative global emissions, followed by a shallowing of the temperature-emissions relationship for fractional shares exceeding 75% (where the relationship reflects the response progressively earlier in the industrial period) (Figure 3g).

4. Discussion

We use a generative diffusion model to quantify the probability that surface temperatures would be altered if the same large-scale meteorological conditions were to occur without a given fractional share of cumulative anthropogenic emissions (with high probabilities corresponding to low overlap between the predicted distributions; Figure 3e and Figure S2 in Supporting Information S1). While it is clearly established that, in aggregate, global greenhouse gas emissions since the pre-industrial period have increased the frequency and intensity of extreme heat events, natural climate variability can make it difficult to detect robust changes in individual events caused by small increments of emissions, highlighting the importance of rigorous uncertainty quantification in this analysis. To achieve this, we specifically design our diffusion model to capture two dominant sources of uncertainty: natural climate variability and GCM-differences in global climate sensitivity and the regional climate response (see Methods). While natural variability presents the dominant source of attribution uncertainty for small changes in emissions and short-duration events, we find that GCM differences become increasingly important for larger emission shares and longer-duration events (Figure 3g and Figure S11 in Supporting Information S1), consistent with the increasing spread in global and regional-mean temperatures between individual GCMs at high emissions (Figure S12 in Supporting Information S1). These results suggest that accurately quantifying both of these sources of uncertainty is crucial for making robust attribution estimates.

Importantly, we find that the probability associated with an attributable temperature change can be strongly dependent on the timescale and region of interest, due to the larger signal-to-noise ratio for longer duration events with larger spatial extents (Figure S11 in Supporting Information S1). For example, although we estimate >99% probability that post-2015 emissions increased regional-mean (~2,500 km) temperatures during the 2025 heatwave (Figure 3f), the influence of post-2015 emissions is not as robust at the scale of individual grid cells (~250 km), with only a few grid cells surpassing 95% probability (Figure 3d). As a result, we note that many of our attribution results are event- and region-specific.

Our cumulative emissions time series includes fossil CO₂ emissions from all sources, but excludes CO₂ emissions from land-use change and non-CO₂ forcing agents including aerosols, which may be important drivers of Europe's temperature trends in the last decade (Persad et al., 2025; Schumacher et al., 2024). With this in mind, we demonstrate that our 2025 heatwave attribution results are robust to the treatment of non-CO₂ forcing in our attribution framework by comparing results from two additional neural networks, one trained using cumulative CO₂-equivalent fossil emissions from three dominant greenhouse gases (CO₂, CH₄, and N₂O), and another trained on GCM simulations without non-greenhouse gas forcings (Figure S9 in Supporting Information S1). We also find that the regional response to increasing cumulative CO₂ emissions is more uniform without non-greenhouse gas forcings for both the source GCM simulations (Figure S13a in Supporting Information S1) and the predictions from models trained without non-greenhouse gas forcings (Figure S13b in Supporting Information S1) (although data availability limits this analysis to only two GCMs; see Text S5 in Supporting Information S1). This suggests that non-greenhouse gas forcings may be responsible for driving the muted response of temperature to early-industrial CO₂ emissions (<1000 GtCO₂) (Figure S12b in Supporting Information S1),

and the corresponding flattening of the learned relationship between cumulative CO₂ emissions and temperature for large fractional shares of total historical emissions (i.e., >75%) since 1850 (Figure 3g).

While our main results appear to be insensitive to the explicit inclusion of non-greenhouse gas forcings, more work is needed to incorporate aerosol forcing into extreme event attribution analyses. This is particularly urgent since future aerosol emissions are highly uncertain yet have the potential to cause rapid shifts in historical and future heatwave trends, especially in industrialized regions like Europe (Persad et al., 2025). To address this gap, our attribution approach could be extended (given sufficient training data) to quantify changes in heatwave intensity caused by emissions of each individual greenhouse gas and aerosol constituent, enabling more accurate storyline-based estimates of anthropogenic contributions to individual extreme weather events. However, this would require the available large ensemble of GCM simulations to report emissions of each individual forcing constituent used in their model runs, including spatiotemporal emission data for constituents that are not well-mixed.

Our approach shows promise as a generalizable attribution framework for linking small increments of emissions to changes in individual extreme weather events. However, extending this analysis to other types of events (e.g., precipitation) in different regions may require a broader selection of meteorological inputs to accurately predict changes in event intensity and more extensive bias-correction to prevent GCM biases from dominating the results. Additionally, the ERA5 temperature fields are not independent of the ERA5 meteorological fields (used as diffusion-model inputs) which creates some endogeneity in the evaluation of those predictions that may become especially important for events that are not as accurately captured by reanalysis (e.g., extreme wet events). Therefore, we further reinforce that careful evaluation of the predictions (including possibly to different observational data sets) is essential to ensure the framework is fit-for-purpose for any particular event and region.

5. Conclusions

To understand the extent to which continued greenhouse gas emissions influence extreme weather events, we develop an event-specific attribution approach which uses a generative diffusion model to predict changes in Europe's recent heat extremes due to the combined effects of anthropogenic forcing associated with incremental changes in cumulative fossil CO₂ emissions. Despite decarbonization efforts since the 2015 UN Paris Agreement, we find strong evidence (>95% probability) that post-2015 emissions can be linked to regional-mean changes in the intensity of the hottest week and the number of extreme heat days for each individual year from 2022 to 2025, including Europe's high-impact June 2025 heatwave (although robust changes at individual grid cells are limited). This work is an important step toward understanding the influence of subsets of historical emissions on individual extreme weather events and their impacts.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

The ERA5 (Hersbach et al., 2023) and ERA5-Land (Muñoz Sabater, 2019) data are available from the Copernicus Climate Change Service Climate Data Store (<https://cds.climate.copernicus.eu/datasets>). The CMIP6 data are available from multiple sources and can be accessed through the Earth System Grid Federation website (<https://esgf-node.llnl.gov/search/cmip6/>). Analysis code (Trok, 2026) is available in a Zenodo archive (<https://doi.org/10.5281/zenodo.20728772>).

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